

German Electricity Demand Forecasting Case Study

Advanced Data Science - Time Series Modelling

Name	Student ID	GitHub Repository
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1. Objective and modelling design:

In this work we apply several forecasting methods to forecast the electricity demand in Germany, ranging from simple benchmarks to statistical, feature-based and neural network models. The objective is to determine whether the models are usable in practice apart from minimizing the error. Electricity demand projections are important because grid operators need to balance generation and consumption, but also to avoid unnecessary spending on reserves and reliability problems.

Data were collected from the hourly time-series file of the Open Power System Data. German real load column, DE_load_actual_entsoe_transparency, was split and filtered from 01.01.2015 until the end of available 2020 records. For the exploratory analysis and benchmark, SARIMA, SARIMAX and Random Forest models, we generated the daily and weekly average series. For the LSTM experiment, we directly used the hourly data. The last 104 weekly observations were held out for a two-year test period.

2. Data preparation and preliminary results:

The load column was renamed to a more descriptive variable name, and the timestamp was transformed into a time-series index. The cleaned working data set included 50,400 hourly, 2,100 daily, and 301 weekly observations. No further missed German load observations after preparation. Weekly averaging was selected for most models, as it preserves the annual seasonal trend, and reduces short-term hourly noise.

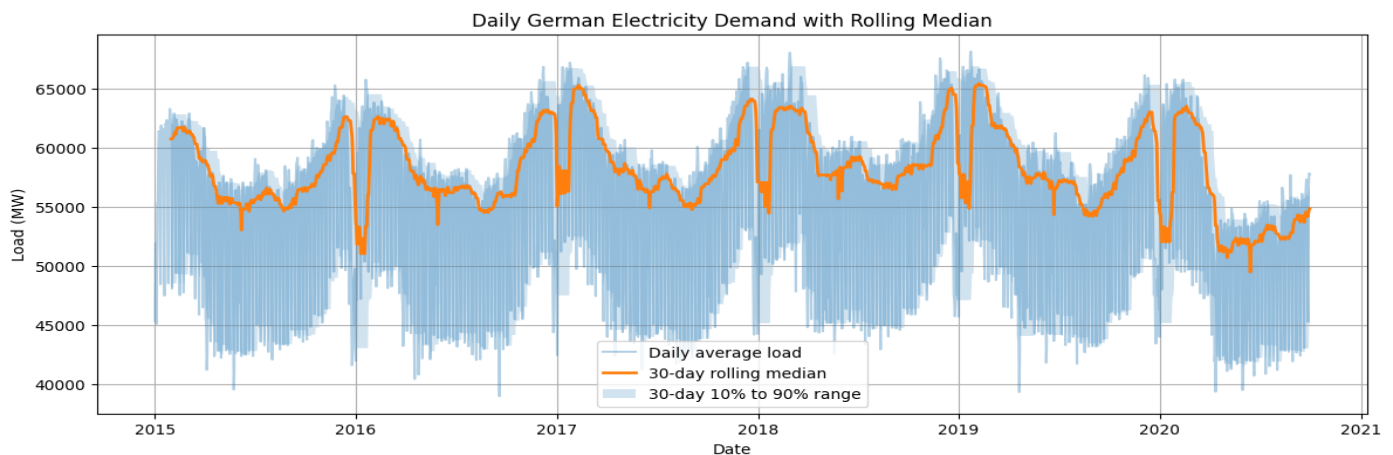


Figure 1. Daily electricity demand with rolling median and variation band.

The exploratory analysis showed that there were annual patterns of activity rather than random movement. Generally, demand was higher in the colder months and lower in warmer ones. The load duration curve shows a high base level of demand for most of the time and fewer high-load times that are important for peak planning. The monthly seasonal indicator suggests that demand levels fluctuate throughout the year.

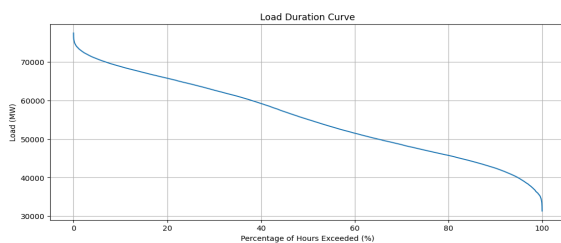


Figure 2a. Load duration curve.

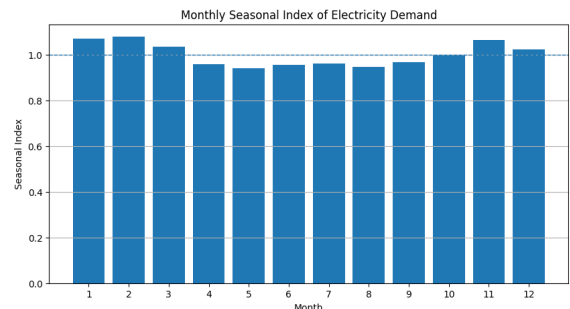


Figure 2b. Monthly seasonal index.

3. Autocorrelation and stationarity:

The ADF and KPSS tests address stationarity from different perspectives, so they were used together. The weekly load was statistically suitable for modelling with a p-value of ADF 0.00118 and a p-value of KPSS 0.100. Moreover, a first difference was tested and provided more convincing evidence of stationarity. However, the seasonal structure was maintained as an important feature of the modelling as the ACF/PACF plots still showed seasonal dependence.

Series	ADF statistic	ADF p-value	KPSS statistic	KPSS p-value	Interpretation
Weekly load	-4.047	0.00118	0.160	0.100	Stationary by both tests
First difference	-7.069	<0.001	0.056	0.100	Stronger stationarity

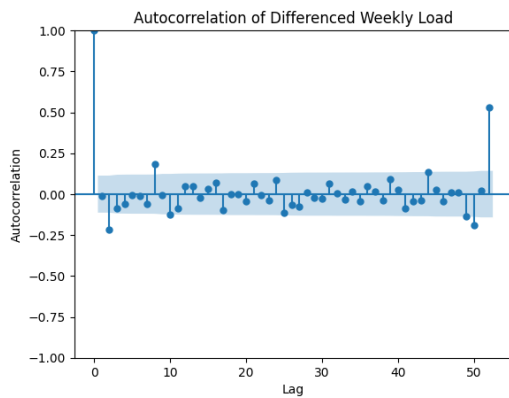


Figure 3a. ACF of differenced weekly load.

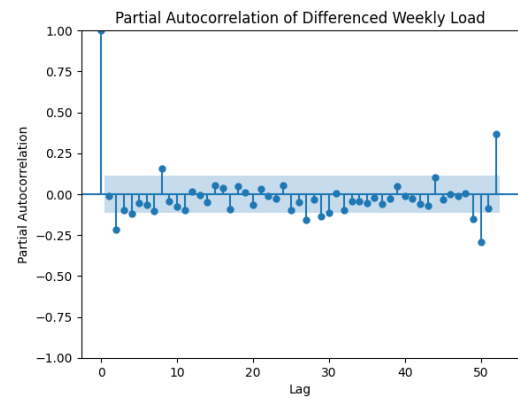


Figure 3b. PACF of differenced weekly load.

4. Forecasts for benchmark's Models:

Prior to the advanced models, four benchmark forecasts, mean, naive, seasonal naive, and drift—were fitted. These are crucial since a sophisticated model can only be valuable if it outperforms a straightforward baseline. The best benchmark was the seasonal naïve approach, which replicates the same week from the prior year.

Benchmark model	RMSE	MAE	MAPE (%)
Seasonal naive	3006.76	2318.52	4.41
Mean forecast	4397.30	3788.83	6.97
Naive forecast	4459.11	3783.20	6.79
Drift forecast	5117.96	4339.89	8.05

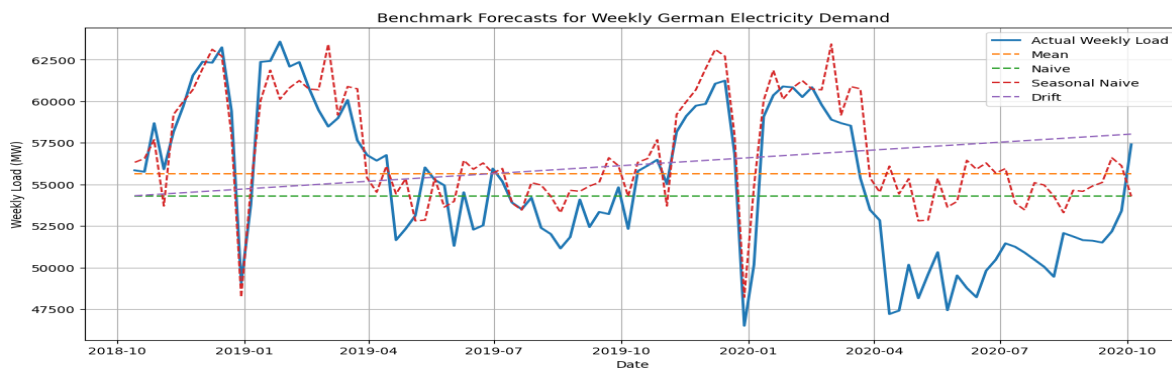


Figure 4. Benchmark forecasts against observed weekly demand in the two-year test period.

The benchmark data shows a very strong yearly recurrence of German power demand. Linear trend was not appropriate because the drift model had the worst performance. Subsequent models used the seasonal naïve outcome as their main benchmark.

5. The SARIMA Model:

The weekly data showed autocorrelation and seasonality over the year, so a SARIMA model was suitable. The seasonal period was chosen to be 52 weeks. The necessary grid search across $p = 0$ to 6, $d = 0$ to 2, and $q = 0$ to 6 found 147 non-seasonal parameter combinations. The final forecast Version 3 was SARIMA (4,2,6) (1,1,1,52). However, the best likelihood choice by AIC is SARIMA (2,2,6) (1,1,1,52).

- The modelling frequency was weekly, where 52 data equaled a yearly cycle. $m=52$, which was likewise compatible with the winter-summer seasonal pattern and the seasonal dependence observed during exploration, was consistent with the seasonal period. The last SARIMA $((4,2,6) (1,1,1,52))$ was non-seasonal differencing $(d=2)$ and seasonal differencing $(D=1)$. The ADF and KPSS evidence revealed that the original series did not need particularly heavy non-seasonal differencing. These ordering arose from the parameter search. The poor out-of-sample performance reinforces this concern: differencing twice may have eliminated useful level information and made the model less sensitive in the unusual timeframe of 2019–2020. A future version should compare the chosen structure with simpler alternatives of $(d=0)$ and $(d=1)$ using rolling validation instead of depending solely on likelihood fit.

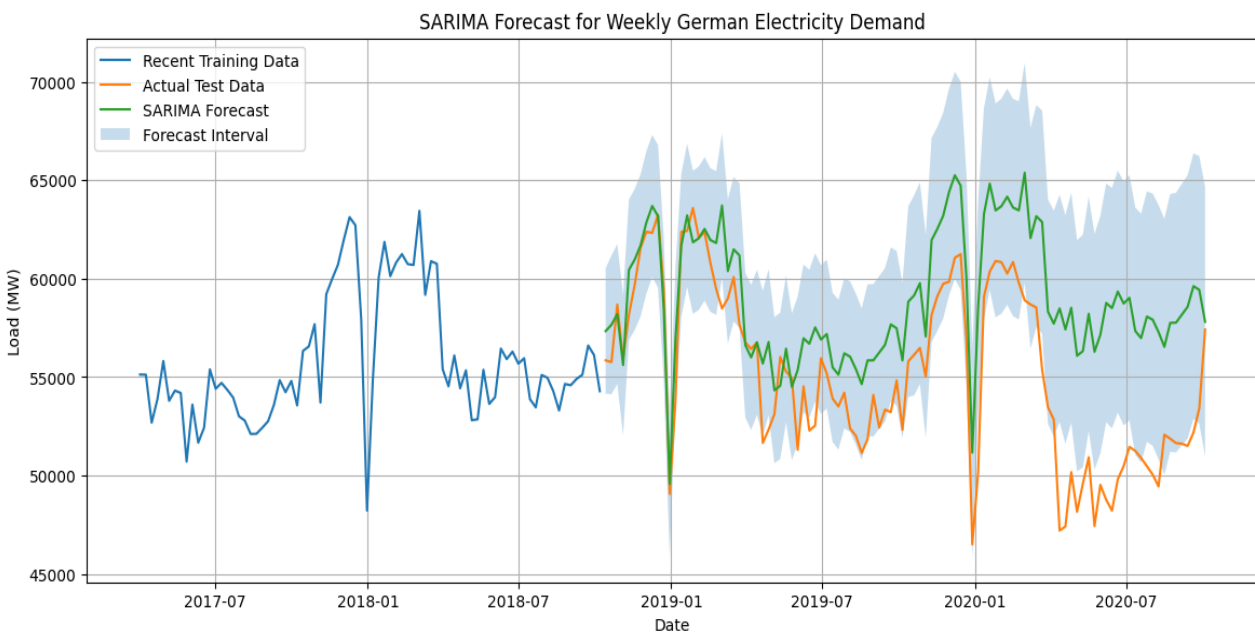


Figure 5. Final SARIMA forecast with confidence interval.

Model	Frequency	RMSE	MAE	MAPE (%)
Seasonal naïve	Weekly	3006.76	2318.52	4.41
SARIMA	Weekly	4765.23	3883.62	7.37

The seasonal naïve benchmark outperformed the SARIMA model. This conclusion is helpful because it demonstrates that a strong hold-out forecast is not always the consequence of a solid likelihood fit. There may have been less generalization during the test time due to over-differencing and atypical behavior. To determine whether structure persisted after modelling, the residual histogram and residual ACF were examined.

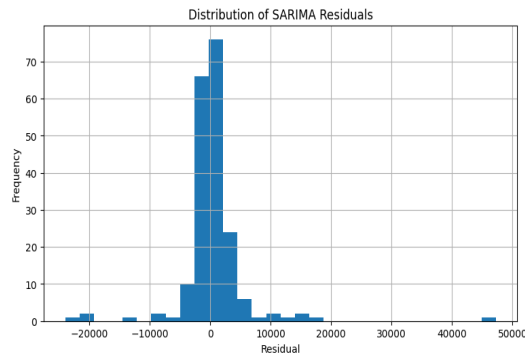


Figure 6a. SARIMA residual distribution.

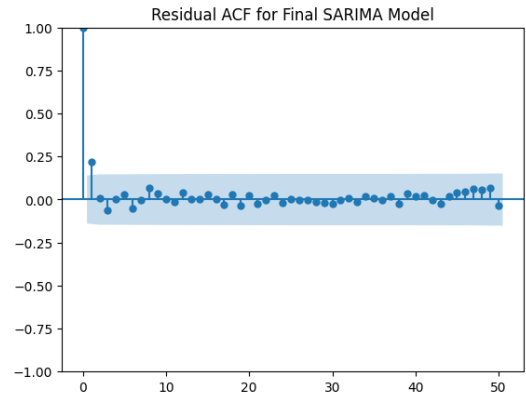


Figure 6b. SARIMA residual ACF.

6. SARIMAX with holiday and weather variables

In a SARIMAX model, the temperature in Berlin and the number of German holidays were included as exogenous variables. Temperature was added since holidays can alter commercial and industrial activity, and heating and cooling behavior might impact load. Although future observed temperatures are unknown at the forecast origin unless they are derived from a weather forecast, holiday variables are known beforehand. As a result, the SARIMAX result should be referred to as an explanatory or conditional forecast.

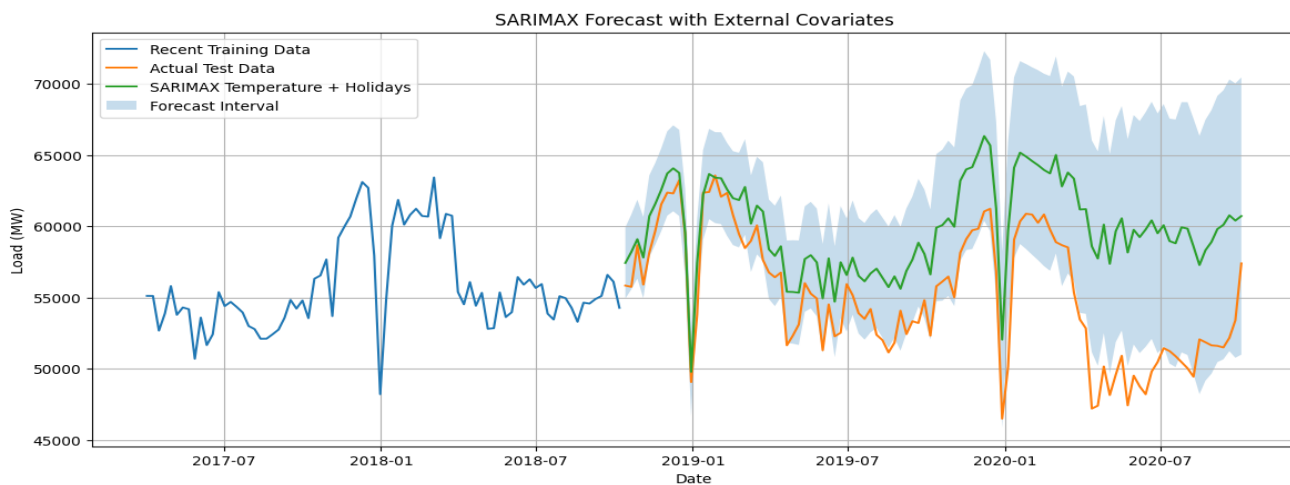


Figure 7. SARIMAX forecast using Berlin temperature and holiday covariates.

Model	Frequency	RMSE	MAE	MAPE (%)
Seasonal naive	Weekly	3006.76	2318.52	4.41
SARIMAX temperature + holidays	Weekly	5633.88	4712.47	8.92

The additional covariates in this version did not increase accuracy. This implies that the SARIMAX structure did not make good use of temperature and holidays, but it does not imply that they are worthless. Additionally, a single Berlin temperature series is merely a stand-in for the national weather of Germany. It would be preferable to use a population-weighted temperature index from multiple cities.

- The numerical comparison verifies the conclusion that the covariates did not increase forecast accuracy in this trial. The RMSE of the Temperature-and-holiday SARIMAX approach was 5633.88 and MAPE was 8.92%, whereas the Seasonal Naive model produced RMSE 3006.76 and MAPE 4.41%. This is not to say that weather and holidays are irrelevant; it rather shows that the SARIMAX structure and the single-city Berlin climate proxy used were not good at making use of them. Holiday dates are known ahead of forecast and hence are operationally safe. The historical and available one-week temperature

lag but without a weather prediction, you wouldn't know the temperature in the same week. Therefore, results based on that variable should be considered conditional or explanatory forecasts.

7. The Random Forest feature model:

A Random Forest regressor with lagged load values, rolling statistics, calendar variables, weather characteristics, and holiday indicators was employed in the feature-based model. Only data that was accessible prior to the projected week was utilised by shifting the lag and rolling characteristics. This was crucial to preventing data leaks. Random Forest does not need the weekly series to be stationary in order to model non-linear relationships.

- The feature table was developed with the protection of the information border. The demand series was lagged to ensure the target week was not one of its own predictors. Load lags and rolling summaries were generated. The variable `temperature_lag1` was also created by changing temperature by one week. The model still kept observed same-week temperature and a temperature-change variable calculated from current and prior temperature. These variables do not leak the goal load, but would not be available in a real prediction unless provided by a weather service. The load and lagged-temperature aspects are thus operationally valid, but outcomes incorporating observed same-week weather are conditional.

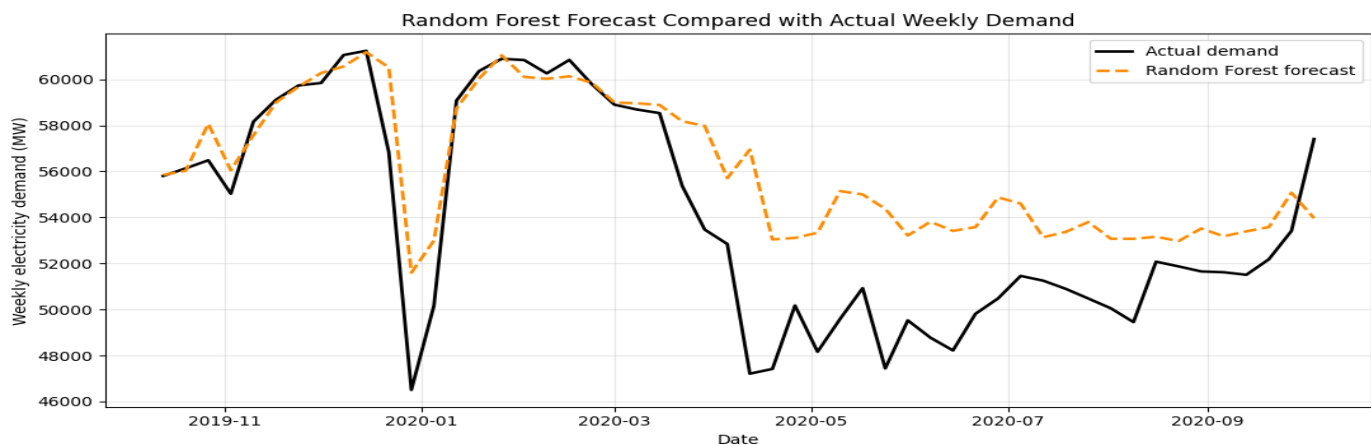


Figure 8. Random Forest weekly forecast against actual demand.

The best weekly model was Random Forest. For the seasonal naïve benchmark, it lowered MAPE from 4.41% to 3.51%. Lagged demand, particularly the 52-week lag, dominated feature relevance, indicating that load from the prior year is a significant predictor of future demand.

Model	Frequency	RMSE	MAE	MAPE (%)
Seasonal naïve	Weekly	3006.76	2318.52	4.41
Random Forest	Weekly	2505.60	1826.49	3.51

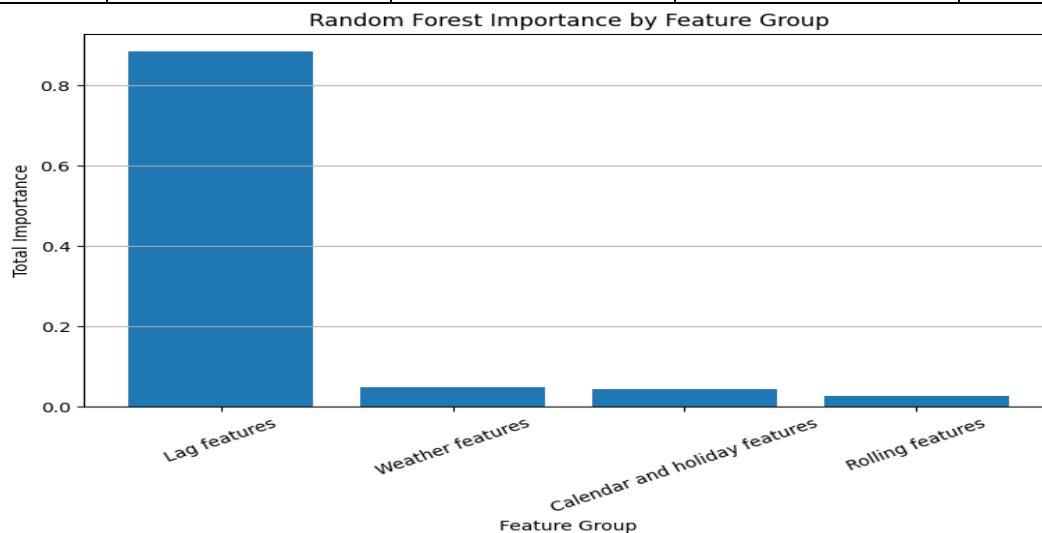


Figure 9. Grouped feature importance from the Random Forest model.

8. The hourly LSTM model:

The initial hourly demand data was used to train the LSTM model. As a result, the model was able to learn weekly and daily temporal patterns that are hidden after weekly aggregate. To prevent leakage from subsequent observations, the scaler was only installed during the training phase and then applied during the test period. Before choosing the final model, a quick tuning check contrasted different lookback and layer settings.

LSTM setting	Hours	Layer design	Dropout	Epochs	RMSE	MAPE (%)
Weekly Memory LSTM	168	50 + 50	0.20	8	1690.02	2.40
Two Week Memory LSTM	336	64 + 32	0.25	4	2247.75	3.17
Short Memory LSTM	48	32	0.20	4	2231.35	3.09

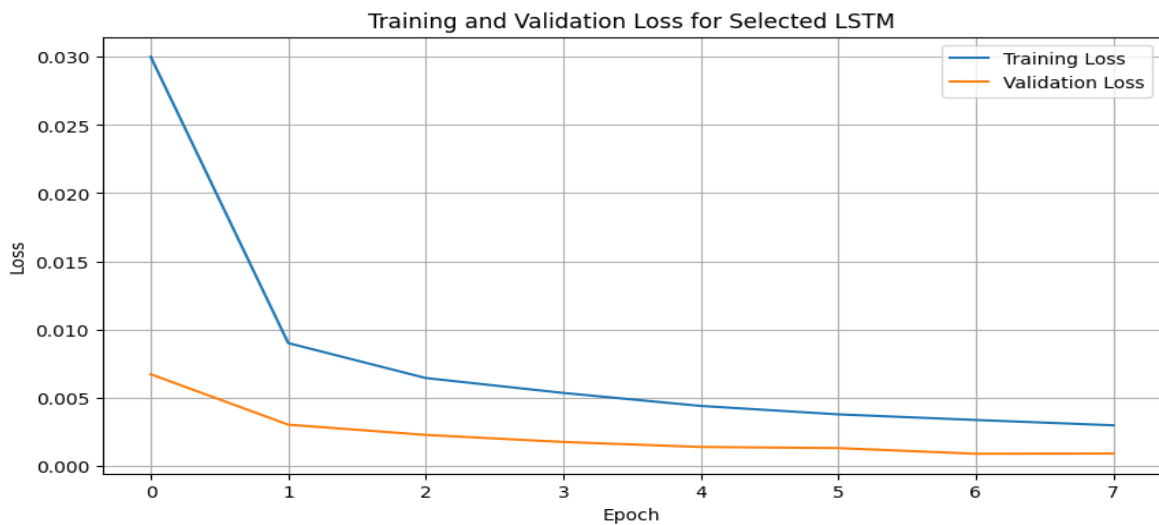


Figure 10. Training and validation loss for the selected LSTM model.

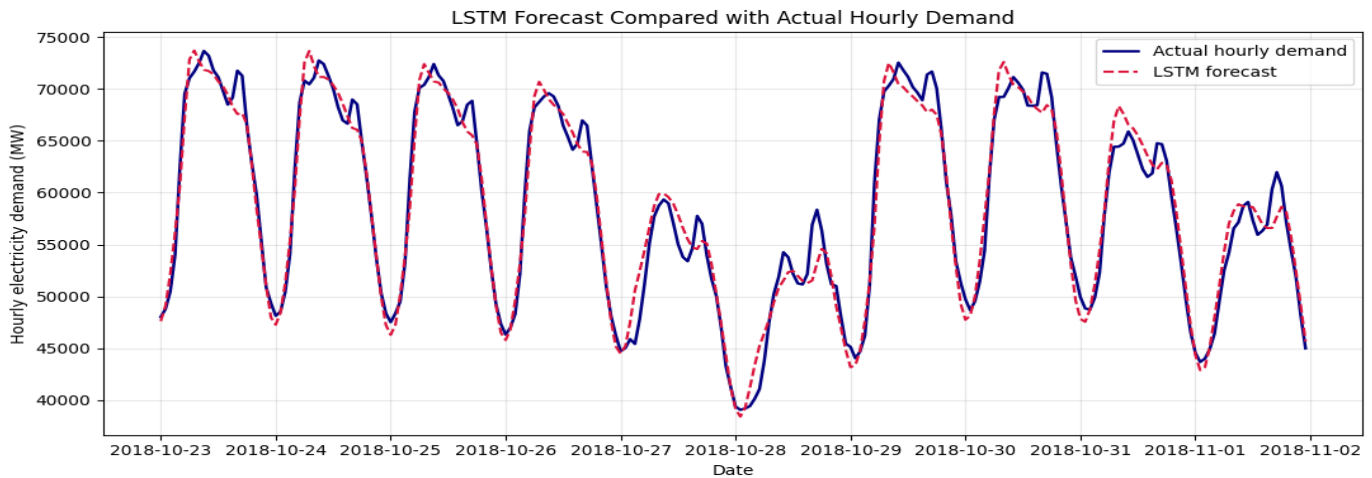


Figure 11. LSTM forecast sample against actual hourly demand

With a MAPE of 2.40%, the LSTM produced the lowest percentage error. SARIMA, SARIMAX, and Random Forest are weekly models, whereas this outcome is assessed using hourly observations. Therefore, if hourly demand forecasting is the operational goal, the LSTM is most pertinent. Compared to the Random Forest model, it is also harder to understand and takes more upkeep.

9. Final comparison and interpretation:

Model	Frequency	RMSE	MAE	MAPE (%)
LSTM	Hourly	1690.02	1282.10	2.40
Random Forest	Weekly	2505.60	1826.49	3.51
Seasonal naive	Weekly	3006.76	2318.52	4.41
SARIMA	Weekly	4765.23	3883.62	7.37
SARIMAX temperature + holidays	Weekly	5633.88	4712.47	8.92

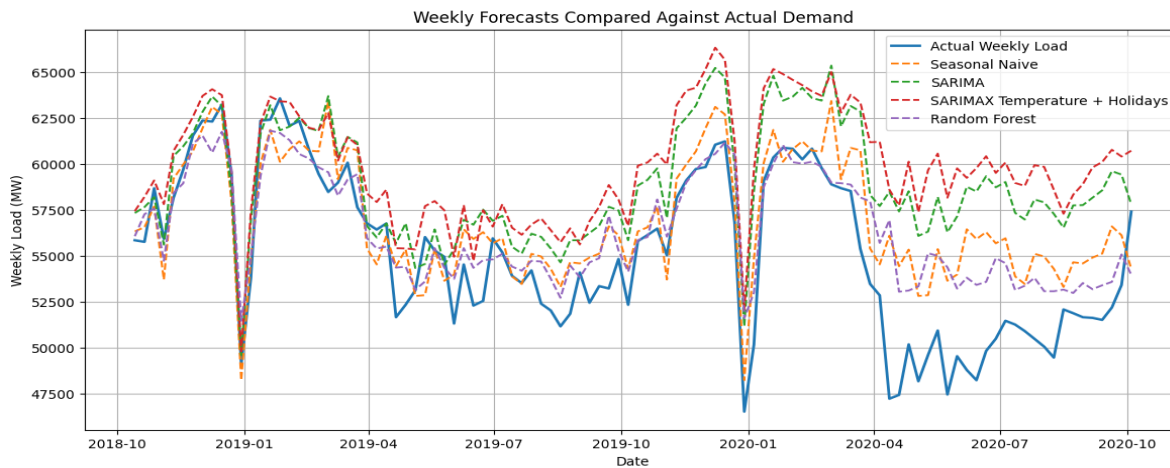


Figure 12. Final weekly forecast comparison

Seasonal Naive set a tough weekly standard with RMSE 3006.76 and MAPE 4.41%. The only significant improvement was in the weekly prediction using the Random Forest method with RMSE=2505.60 and MAPE=3.51%. This is a reduction in RMSE of around 16.7% which indicates that lagged demand, calendar information and engineering features contributed value beyond just repeating the corresponding week from the prior year. Benchmark could not be beaten by SARIMA with RMSE 4765.23 and MAPE 7.37% and SARIMAX with RMSE 5633.88 and MAPE 8.92%. The LSTM achieved an RMSE of 1690.02 and a MAPE of 2.40%, although these values were derived from hourly observations. The errors should not be compared directly with weekly results without first aggregating both forecasts and observations to the same frequency.

There were also major differences in transparency and upkeep between the models. SARIMAX proved to be the easiest advanced model to describe, as there were clear statistical functions for the autoregressive, moving-average,

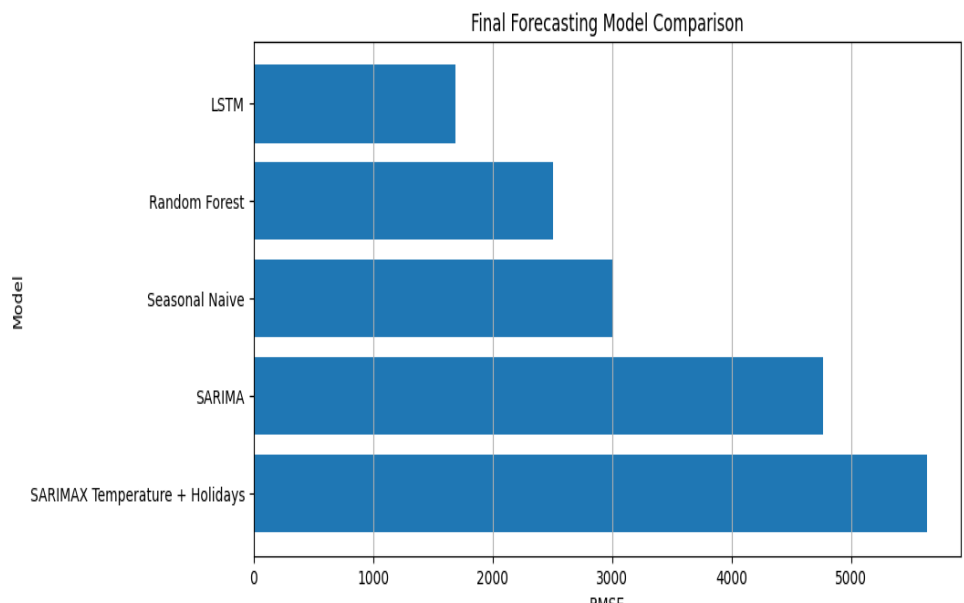


Figure 13: Models Comparison Bar graph

differencing, seasonal and exogenous parts. It also provided forecast intervals to communicate uncertainty. Random Forest was moderately interpretable. Individual tree decisions were less clear, but feature importance showed which inputs affected the forecast. It also dealt with non-linear interactions without the same stationarity assumptions. LSTM was the most sophisticated and less transparent model due of the need of scaling, sequence construction, architectural adjustment and extra computational monitoring. Its hidden-state representation also made the individual forecast difficult to communicate to a non-technical person.

10. Limitations, conclusion, and future work:

The study has shown that there is a strong yearly and calendar-related pattern in German electricity demand. The seasonal naive standard was hard to beat, indicating the importance of demand in previous years. The best hourly result was from the LSTM model while the best weekly model was the Random Forest model. For this Version 3 experiment SARIMA and SARIMAX did not generalise as well as we would have hoped, though they were useful for interpretation and uncertainty.

The Random Forest is the most practicable equilibrium for weekly operational use. It performed much better on accuracy than Seasonal Naive, was simpler to retrain and maintain than the LSTM, and provided feature-importance evidence that could be presented to users. Its biggest issue is uncertainty, as it doesn't automatically give confidence intervals like SARIMAX does. Thus, before deployment, we need to add bootstrap, quantile or conformal intervals. The operational model should use lagged demand, calendar variables and known holidays. Any input on the current temperature should come from an actual weather forecast. Seasonal Naive should be kept as a simple backup and monitoring benchmark, as it is stable, cheap and easy to update.

The main limitations are the fixed train-test split, the use of the temperature of Berlin as a proxy of the national one, and the difficulty of predicting the 2019-2020 period from the past data. Future research should include multiple German meteorological stations, rolling-origin cross-validation, probabilistic machine learning forecasts and other neural architectures such as CNN-LSTM or GRU.

12. References:

1. Analysis of Time Series: Forecasting and Control Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015). Hoboken: Wiley, 5th ed. P. 9781118675021 TIME SERIES EVALUATION FORECASTING AND CONTROL 5TH EDITION, GENERAL INTRODUCTORY STATISTICS, Wiley
<https://www.wiley.com/en-us/shop/general-introductory-statistics/time-series-analysis-forecasting-and-control-5th-edition-p-9781118675021>
2. Dickey, D. A. "Distribution of Estimators for Autoregressive Time Series with a Unit Root. and Fuller, W. A. Journal of the American Statistical Association 74(366a): 427-431. (1979).
<https://www.tandfonline.com/doi/abs/10.1080/01621459.1979.10482531>
3. "Testing the null hypothesis of stationarity against the alternative of a unit root." Kwiatkowski, D., Phillips, P.C.B., Schmidt, P. and Shin, Y. (1992). Journal of Econometrics, 54, 159–178. Use this for the KPSS test. This is the original document for KPSS Stationarity Test.
https://econpapers.repec.org/article/eeeeconom/v_3a54_3ay_3a1992_3ai_3a1-3_3ap_3a159-178.htm
4. Breiman, L. Random forests. (2001) Machine Learning, 45, pp. 5-32. This is for your feature based Random Forest model. This is the original Random Forest paper, a trusted academic source.
<https://link.springer.com/article/10.1023/A:1010933404324>
5. Hong, T. "Probabilistic electric load forecasting: A tutorial review," International Journal of Forecasting 32, 3 (2016): 914-938. and S. Fan. (2016). Use it to explore operational forecasting, uncertainty and the context of electrical load forecasting. This evaluation of load forecasts shall be subject to peer review.
<https://research.monash.edu/en/publications/probabilistic-electric-load-forecasting-a-tutorial-review/>
6. Fan, S. and Hyndman, R. J. (2012). 'Short-term load forecasting based on a semi-parametric additive model', IEEE Transactions on Power Systems, 27(1), pp. 134–141.
<https://doi.org/10.1109/TPWRS.2011.2162082>